Chapter 6 — Heat and Thermodynamics

Full textbook exercise · PECTAA Physics 11 · No ALP filtering

Q.1 — Multiple Choice Questions (with correct option and reason)

6.1. In an isothermal change, internal energy:

(a) decreases **(b)** increases
(c) remains the same **(d)** becomes zero

Answer: (c) remains the same — For an ideal gas, U depends only on temperature.

6.2. First law of thermodynamics is based upon conservation of:

(a) mass **(b)** energy
(c) momentum **(d)** charge

Answer: (b) energy — It is energy conservation for thermal systems.

6.3. A device which converts thermal energy into mechanical energy is called:

(a) heat engine **(b)** Carnot engine
(c) refrigerator **(d)** turbine

Answer: (a) heat engine — General term is heat engine.

6.4. Two objects in thermal contact with same temperature are at:

(a) thermal equilibrium **(b)** chemical equilibrium
(c) mechanical equilibrium **(d)** physical equilibrium

Answer: (a) thermal equilibrium — No net heat flows.

6.5. When a system expands by adding heat, the work done is:

(a) positive and on the system **(b)** negative and on the system
(c) positive and by the system **(d)** negative and by the system

Answer: (c) positive and by the system — Expansion work is done by the system.

6.6. Entropy of a system in a reversible process:

(a) decreases **(b)** increases
(c) is infinite **(d)** is zero

Answer: (d) is zero — Total entropy change is zero for reversible process.

6.7. What happens to internal energy of an object when its temperature increases?

(a) decreases **(b)** remains constant
(c) increases **(d)** fluctuates

Answer: (c) increases — Higher temperature means greater molecular kinetic energy.

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6.8. The value of Boltzmann constant is:

(a) 1.38×10^{-23} J K^{-1} **(b)** 1.38×10^{23} J K^{-1}
(c) 1.38×10^{-23} K^{-1} **(d)** 1.38×10^{23} K^{-1}

Answer: (a) 1.38×10^{-23} J K^{-1} — $k_B = R/N_A$.

6.9. In an adiabatic process, there is no:

(a) change in temperature **(b)** exchange of heat
(c) change in internal energy **(d)** work done

Answer: (b) exchange of heat — Q = 0.

6.10. Thermodynamics mostly deals with:

(a) measurement of quantity of heat **(b)** transfer of quantity of heat
(c) changes of state **(d)** conversion of heat to other forms of energy

Answer: (d) conversion of heat to other forms of energy — It studies heat, work and energy conversion.

Q.2 — Short Answer Questions

6.1. What is meant by thermal equilibrium?

Two bodies are in thermal equilibrium when they are at the same temperature and there is no net heat flow between them.

6.2. What is internal energy? How is it related to temperature of an ideal gas?

Internal energy is the total kinetic and potential energy of molecules. For an ideal gas, potential energy is negligible, so internal energy depends only on temperature and increases with it.

6.3. State the second law of thermodynamics in two different forms.

Kelvin–Planck: no cyclic engine can convert all absorbed heat into work without rejecting some heat. Clausius: heat cannot flow from cold to hot without external work.

6.4. Is it possible to construct a heat engine of 100% efficiency?

No. It would require rejecting no heat, which would need a cold reservoir at 0 K. The second law forbids complete conversion of heat into work in a cycle.

6.5. Differentiate between reversible and irreversible processes.

A reversible process can be retraced exactly with no net change in system or surroundings. An irreversible process cannot be exactly reversed and increases total entropy.

Feature	Reversible	Irreversible
Speed	Infinitely slow	Finite/fast

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Feature	Reversible	Irreversible
Entropy	Total change zero	Total entropy increases
Example	Ideal Carnot cycle	Free expansion, friction

6.6. Why is an adiabatic curve steeper than an isotherm?

In adiabatic expansion temperature also falls, so pressure drops faster than in isothermal expansion where temperature is constant.

6.7. A refrigerator transfers heat from cold to hot. Does this violate the second law?

No. External work is supplied by the compressor, so the transfer is allowed and total entropy increases.

6.8. Explain heat death of the universe in terms of entropy.

If entropy reaches a maximum, energy becomes uniformly spread and no temperature differences remain. No further work can be extracted, so the universe reaches a dead thermal state.

6.9. Can a cyclic reversible engine absorb heat at constant temperature and convert it completely into work without rejecting heat?

No. Even a reversible engine must reject some heat to a colder reservoir; otherwise it would violate the second law.

6.10. How do real gases differ from ideal gases at high pressure and low temperature?

At high pressure molecular volume is not negligible; at low temperature intermolecular attractions become important. Both cause deviations from PV = nRT.

6.11. Show that area under a P–V graph equals work done.

For a small expansion, W = PΔV, the area of a narrow strip. Summing all strips gives total work, equal to the area under the curve.

6.12. How is work done (i) by a gas (ii) on a gas?

(i) During expansion the gas does positive work on the surroundings. (ii) During compression work is done on the gas, taken as negative in the convention Q = ΔU + W_by.

Constructed Response Questions

6.1 Explain how thermodynamics relates to energy conservation.

The first law states Q = ΔU + W: heat supplied is used to increase internal energy and do work. Energy is neither created nor destroyed, only transferred or converted.

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6.2 Explain how thermodynamics applies to biological systems such as the human body.

The body takes in chemical energy from food, converts part into work and heat, and maintains temperature. It obeys the first law; metabolism and heat loss are thermodynamic processes.

6.3 A gas expands adiabatically. What happens to temperature and pressure?

No heat enters, so work done comes from internal energy. Temperature falls, and pressure falls more rapidly than in isothermal expansion.

6.4 A coffee cup cools on a table. Explain the thermodynamic processes.

Heat flows from hot coffee to cooler surroundings by conduction, convection and radiation until thermal equilibrium is reached. Entropy of the surroundings increases more than the coffee's entropy decreases.

6.5 How can weather patterns such as wind and rain be explained by thermodynamics?

Uneven heating creates pressure differences, causing wind. Rising warm air expands and cools adiabatically, leading to condensation, clouds and rain.

Comprehensive / Long Questions

6.1 What are the postulates of kinetic theory? Derive PV = Nk_BT from the general gas equation.

Kinetic theory assumes large numbers of tiny particles in random motion, negligible molecular volume, no forces except during elastic collisions, and average kinetic energy proportional to T. From PV = nRT and n = N/N_A, PV = N(R/N_A)T = Nk_BT.

6.2 State and explain various gas laws.

Boyle's law: at constant T, $P \propto 1/V$. Charles' law: at constant P, $V \propto T$. Gay-Lussac's law: at constant V, $P \propto T$. Together they give PV/T = constant for a fixed mass of ideal gas.

6.3 Explain the first law of thermodynamics in detail. Give an example and two applications.

$Q = \Delta U + W$. Example: heating a gas in a cylinder increases internal energy and pushes a piston. Applications: isothermal process ($\Delta U = 0$, $Q = W$) and adiabatic process ($Q = 0$, $\Delta U = -W$).

6.4 What is a refrigerator? Explain working and derive coefficient of performance.

A refrigerator uses external work to extract heat Q_2 from a cold reservoir and reject $Q_1 = Q_2 + W$ to a hot reservoir. $COP = Q_2/W = Q_2/(Q_1 - Q_2)$.

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6.5 What is a Carnot engine? Describe the cycle, state Carnot theorem and derive efficiency.

A Carnot engine is an ideal reversible engine with two isothermal and two adiabatic steps. Carnot theorem: no engine between the same reservoirs is more efficient. Efficiency $\eta = 1 - Q_2/Q_1 = 1 - T_2/T_1$.

6.6 Define and explain entropy.

Entropy is a state variable measuring disorder. For reversible heat transfer, $\Delta S = \Delta Q/T$. In natural processes total entropy increases, and energy becomes less available for work.

Numerical Problems (fully worked)

6.1 A gas occupies 6.0 L at 12 atm. Pressure is increased by 2.0 atm at constant temperature. Find new volume.

Given: $P_1 = 12$ atm, $V_1 = 6.0$ L, $P_2 = 14$ atm

Formula: $P_1V_1 = P_2V_2$

$V_2 = 12\times6/14 \approx 5.14$ L.

Answer: ≈ 5.14 L

The book prints 6.14 L; using $P_2 = 14$ atm gives 5.14 L.

6.2 A vacuum chamber has pressure 1.00 nPa at 300 K. Find number of molecules in 1.00 m³.

Given: $P = 1.00\times10^{-9}$ Pa, $V = 1.00$ m³, $T = 300$ K

Formula: PV = Nk_BT

$N = PV/(k_BT) = 10^{-9}/(1.38\times10^{-23}\times300) \approx 2.41 \times 10^{11}$ molecules.

Answer: 2.41×10^{11} molecules

6.3 A gas absorbs 500 J and performs 300 J work. Find change in internal energy.

Given: $Q = 500$ J, $W = 300$ J

Formula: $\Delta U = Q - W$

$\Delta U = 500 - 300 = 200$ J.

Answer: 200 J

6.4 A Carnot engine operates between 600 K and 300 K. Find (i) maximum efficiency (ii) work output if it absorbs 500 J.

Given: $T_1 = 600$ K, $T_2 = 300$ K, $Q_1 = 500$ J

Formula: $\eta = 1 - T_2/T_1$, $W = \eta Q_1$

$\eta = 1 - 300/600 = 0.5 = 50\%$.

$W = 0.5 \times 500 = 250$ J.

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Answer: (i) 50% (ii) 250 J

6.5 A refrigerator extracts 1200 J and releases 1800 J per cycle. Find (i) work input (ii) COP.

Given: $Q_2 = 1200$ J, $Q_1 = 1800$ J

Formula: $W = Q_1 - Q_2$, $COP = Q_2/W$

$W = 1800 - 1200 = 600$ J.

$COP = 1200/600 = 2$.

Answer: (i) 600 J (ii) 2

6.6 Calculate entropy change when 1.0 mole of ice at 0°C melts. Heat of fusion per mole = 6.01×10^3 J.

Given: $Q = 6.01 \times 10^3$ J, $T = 273$ K

Formula: $\Delta S = Q/T$

$\Delta S = 6010/273 \approx 22.0$ J K^{-1}.

Answer: 22.0 J K^{-1}

6.7 A gas occupies 400 mL at 20°C. Find volume at 80°C at constant pressure.

Given: $V_1 = 400$ mL, $T_1 = 293$ K, $T_2 = 353$ K

Formula: $V_1/T_1 = V_2/T_2$

$V_2 = 400 \times 353/293 \approx 482$ mL.

Answer: 482 mL

6.8 A gas has pressure 2 atm at 300 K. Find pressure at 450 K at constant volume.

Given: $P_1 = 2$ atm, $T_1 = 300$ K, $T_2 = 450$ K

Formula: $P_1/T_1 = P_2/T_2$

$P_2 = 2 \times 450/300 = 3$ atm.

Answer: 3 atm

Textbook checks flagged in this chapter

• Numerical 6.1: book prints 6.14 L; with pressure increased from 12 atm to 14 atm, the result is 5.14 L.

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